Himanshu Kumar

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Bachelor of Technology
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GitHub | LinkedIn
LeetCode | CodeChef

Education

•Bachelor of Technology in Information Technology *2022-26*
Rajiv Gandhi Institute of Petroleum Technology, Hoskote, Bengaluru CGPA: 8.36

Research & Publication | [Link]

•A DEC Method for Location-Specific Driving Safety Profiling using Trajectory Data *2025*
Manuscript accepted *Nature Journal*

- Contributed as **Author** (Kushwaha, A. K., **Kumar, H.**, & Kumar, B. A.) by implementing algorithmic refinements, re-running experiments, and addressing reviewer comments to significantly improve manuscript quality.

Work Experience

•Indian Institute of Technology Patna | [Certificate] *May - Jul 2025*
Data Science Intern Patna, India

- Extracted and processed sensor data (LiDAR, accelerometers) from 2,200+ drivers, building a driver behavior model focused on speed, acceleration, and direction-based features.
- Enhanced driver behavior model through implementation of scaling and oversampling techniques; selected and trained K-Means, Mean Shift, DBSCAN and DEC clustering models with optimized parameters.
- Improved model performance using GridSearchCV(Hyperparameter tuning), and clustering metrics (Silhouette, Elbow, CHI, DBI); successfully predicted driver traits namely Aggressive, Speedy and Calm

Projects

•AI-Driven Satellite-Based Emergency Landing Site Prediction for Aircraft | | [Certificate]
Developed an AI-powered model for emergency landing site prediction, using satellite imagery and deep learning.

- Dataset & Processing: Trained on 6362 satellite images using Google Earth APIs and Keras for preprocessing, ensuring efficient feature extraction within 10GB of data.
- Model & Tools Used: Implemented CNNs using TensorFlow for classification and segmentation, with Naive Bayes for risk assessment, running efficiently on a personal laptop (16GB RAM, mid-range GPU).
- Performance & Outcome: Applied data augmentation techniques, improving model accuracy from 78% to 87%, reducing overfitting, and achieving reliable emergency landing site predictions with a 2 hour training time.

•WebSocket-based Real-time Chat Application |
Prepared a WebSocket-based real-time chat application enabling seamless bi-directional communication.

- Architected a WebSocket-based real-time messaging platform, enabling instant communication with a 0.2s delay.
- Introduced real-time user presence indicator and message delivery notifications, reducing response time by 25%
- Designed and implemented a scalable real-time messaging system using Node.js and MongoDB, supporting low-latency bi-directional communication with optimized message handling and improved user engagement by 25%..

Technical Skills and Interests

Languages/Frameworks:C++,Python,HTML5, CSS3, JavaScript, ReactJS (**Frontend**)
Database/Server: MongoDB, MySQL, NodeJS, ExpressJS (**Backend**)
Libraries/Tools: pandas, NumPy, Matplotlib, Github, Jupyter
Relevant Coursework: Data Structures & Algorithms, Operating Systems, OOPs, DBMS, Software Engineering.
Areas of Interest: Web Design and Development, AI/ML.

Achievements

- Springer Nature–accepted research paper (2025)
- Awarded a prestigious sponsorship by HPCL to attend **India Energy Week (IEW) 2026.**
- Finalist at **IIM Bangalore Ideathon 2.0** and Pre-finalist in the **IIT BHU DPBH [Certificate]** *2024*
- **JEE Advanced** Among Top 2 percent of the country *2022*
- **LeetCode and CodeChef** Solved **300**+ DSA Problems *2023-25*

Positions of Responsibility

- **Training and Placement Coordinator** Organized and led recruitment activities for the CSE department.
- **Teaching Assistant**, mentored 80+ students in learning Data Structures and Algorithms (DSA).